PriorCLIP: Visual-Prior-Guided Vision–Language Model for Remote Sensing Image–Text Retrieval

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***Abstract*—Remote sensing image–text retrieval (RSITR) plays a crucial role in remote sensing interpretation, yet remains challenging under both *closed-domain* and *open-domain* scenarios due to semantic noise and domain shifts. To address these challenges within a coherent framework, we propose PriorCLIP, a visual-prior-guided vision–language model that organizes closed-domain and open-domain retrieval as two complementary realizations of the same learning principle. PriorCLIP first introduces remote sensing scene knowledge as a visual prior, then uses this prior to adapt image and text representations to the available training regime, and finally regularizes the representations in a discriminative common space. In closed-domain retrieval, this principle is instantiated at the sample level: the spatial progressive attention encoder (S-PAE) constructs an instruction-conditioned belief matrix to reweight visual tokens and suppress noisy regions, while the temporal progressive attention encoder (T-PAE) progressively propagates contextual information to strengthen text representations. In open-domain retrieval, the same principle is extended to representation transfer through a two-stage strategy that first pretrains image and text encoders on large-scale coarse-grained pairs and then performs vision-instruction fine-tuning on fine-grained pairs. A cluster-based symmetric contrastive affiliation loss further exploits scene-category structure to improve intraclass compactness and interclass separation, thereby reducing semantic confusion in the common space. Extensive experiments on RSICD and RSITMD benchmarks demonstrate that PriorCLIP achieves substantial improvements, outperforming existing methods by 4.9% and 4.0% in closed-domain retrieval, and by 7.3% and 9.4% in open-domain retrieval, respectively.**

***Index Terms*—Image–text retrieval, remote sensing, vision–language models.**

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I. Introduction

REMOTE sensing image–text retrieval (RSITR) is a critical research task that utilizes image patches (or text queries) to retrieve their corresponding textual (or visual) counterparts from large-scale remote sensing databases acquired by satellites or aerial platforms, thereby facilitating the extraction of valuable semantic information [1], [2]. Due to its significant role in practical applications such as natural resource exploration, disaster prevention, and environmental monitoring [3], [4], [5], [6], RSITR has attracted increasing attention from both academia and industry. In recent years, the explosive growth of remote sensing data [7] and the rapid development of cross-modal retrieval methodologies [8], [9] have jointly provided unprecedented opportunities, enabling RSITR to achieve enhanced efficiency, scalability, and accuracy.

According to different training paradigms, RSITR can be divided into closed-domain RSITR (CD-RSITR) and open-domain RSITR (OD-RSITR), as shown in Fig. 1(a). CD-RSITR is a supervised training and retrieval method that operates on a single dataset; however, it is less capable of retrieving more complex ground elements. In closed-domain retrieval, most existing methods utilize CNN-based [10] vision representations and RNN-based [11] text representations, employing pairwise triplet loss (TL) [12] for optimization. Abdullah et al. [13] first applied modality fusion to solve the remote sensing image–text matching problem. Lv et al. [14] utilized knowledge distillation to transform the fusion information of modalities into representations for vision and text learning. Mi et al. [15] utilized a knowledge graph to enrich text semantics and to reduce the information gap between vision and language. Yuan et al. [16] developed an image–text retrieval model to understand multilevel visual semantics, leveraging both global and local information. Pan et al. [17] proposed a scene-aware aggregation network to improve the fine-grained perception of the scene. These approaches benefit from a clear vision and text representation, but perform poorly for long-range dependency modeling [18], [19]. OD-RSITR is fine-tuned on a small dataset after pretraining on an additional large-scale dataset, resulting in improved ground element recognition. CLIP [20] inspired efforts with powerful zero-shot learning on visual interpretation tasks in open-domain retrieval. Subsequently, RemoteCLIP [21], GeoRSCLIP [22], and SkyCLIP [23] models were introduced to establish a foundation vision–language model for remote sensing (RSVLM) and improve large-scale open-domain image–text retrieval

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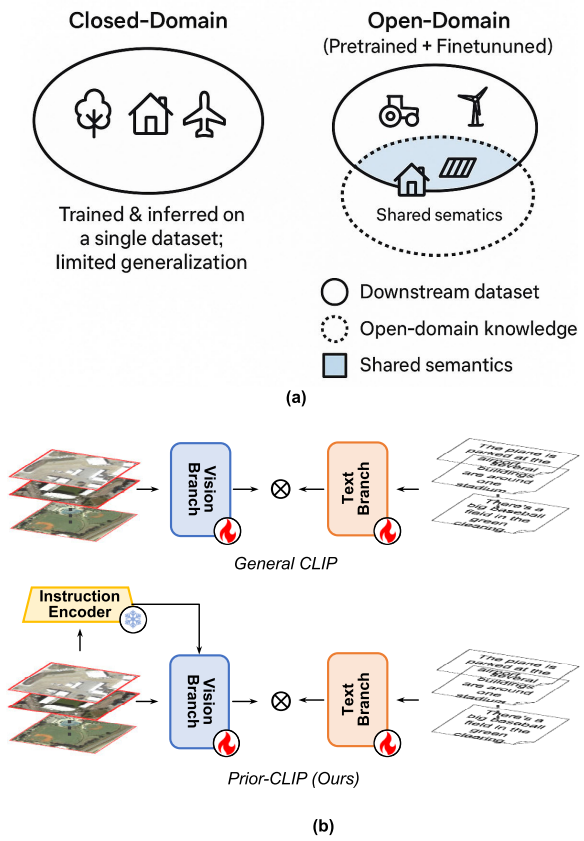


Fig. 1. (a) CD-RSITR and OD-RSITR. (b) Unlike general CLIP structures, PriorCLIP leverages visual priors to guide the image–text alignment process.

performance. Their objective is to develop a foundation of RSVLM, rather than exploring methods to enhance the visual representation of remote sensing.

Despite these advances, RSITR faces challenges that are distinct from those in natural image–text retrieval. Remote sensing images often contain small-scale objects that are more susceptible to semantic noise [16], such as cluttered backgrounds or irrelevant objects. Excessive sensitivity to such noise leads to biased representations and semantic confusion zones [17], i.e., overlapping regions in the shared embedding space that degrade retrieval accuracy. Efforts to address this problem include redundant feature filtering [24] and scene-aware aggregation [17]. However, most approaches still rely on CNN- and RNN-based encoders, which are ineffective for capturing long-range dependencies and scaling with growing data volumes [25]. With the emergence of Transformer-based models in computer vision [26] and natural language processing [27], Transformer encoders for vision and text have shown promise in RSITR. For example, Zhang et al. [28] designed a Transformer-based decoupling module to obtain invariant features across modalities. However, these methods often neglect the unique properties of remote sensing imagery [16], [17], resulting in limited performance gains. Therefore, a fundamental question arises: *how can we design an RSVLM tailored to the specific challenges of RSITR?*

In this article, we propose a visual-prior-guided vision–language model, **PriorCLIP**, to address semantic noise and domain shift through a unified prior-to-representation learning process. As illustrated in Fig. 2, PriorCLIP does not simply assemble separate modules for closed-domain and open-domain retrieval. Instead, both settings share three functional stages: extracting domain knowledge from remote sensing data, using that knowledge to adapt modality-specific representations, and structuring the resulting embeddings in a common semantic space. The distinction between the two settings lies in how this shared principle is instantiated under different data conditions.

For closed-domain retrieval, PriorCLIP performs sample-level prior-guided refinement. A scene-recognition encoder [29] produces an instruction embedding that supplies domain-relevant visual knowledge. The spatial progressive attention encoder (S-PAE) compares this instruction with visual tokens to construct a belief matrix, which emphasizes reliable regions and suppresses background clutter or irrelevant objects. In parallel, the temporal progressive attention encoder (T-PAE) propagates information from preceding textual states to subsequent states, enabling long-range dependency modeling and more coherent sentence representations. The two encoders therefore refine the visual and textual modalities in complementary ways before cross-modal alignment.

For open-domain retrieval, PriorCLIP extends prior-guided adaptation from individual samples to large-scale representation transfer. The image and text encoders are first pretrained on approximately five million coarse-grained remote sensing image–text pairs to acquire broad scene and vocabulary coverage. They are then adapted on fine-grained target pairs through vision-instruction fine-tuning, allowing domain knowledge to refine concept boundaries while retaining the generalization learned during pretraining. Across both settings, a cluster-based symmetric contrastive affiliation loss complements pairwise alignment by exploiting scene-category structure. It encourages image and text embeddings to approach the center of their affiliated category while separating different categories, thereby reducing semantic confusion zones in the common subspace. Extensive experiments demonstrate consistent gains on mAP for both closed- and open-domain retrieval, with ablations validating the contributions of S-PAE and T-PAE, belief-based filtering, the two-stage prior learning strategy, and the proposed affiliation loss. Our PriorCLIP achieves improvements in the CD-RSITR of 4.9% and 4.0% on RSICD [7] and RSITMD [24], respectively, and outperforms open-domain methods by 7.3% and 9.4%. Our key contributions are as follows.

1) We formulate **PriorCLIP** as a unified visual-prior-guided framework in which closed-domain and open-domain retrieval share the same progression from domain-prior extraction to representation adaptation and common-space regularization.
2) For closed-domain retrieval, we instantiate prior-guided adaptation at the sample level through S-PAE and T-PAE. S-PAE uses an instruction-conditioned belief matrix to suppress noisy visual tokens, whereas T-PAE progressively models long-range textual dependencies.
3) For open-domain retrieval, we instantiate the same principle at the transfer level through coarse-grained large-scale pretraining followed by fine-grained vision-

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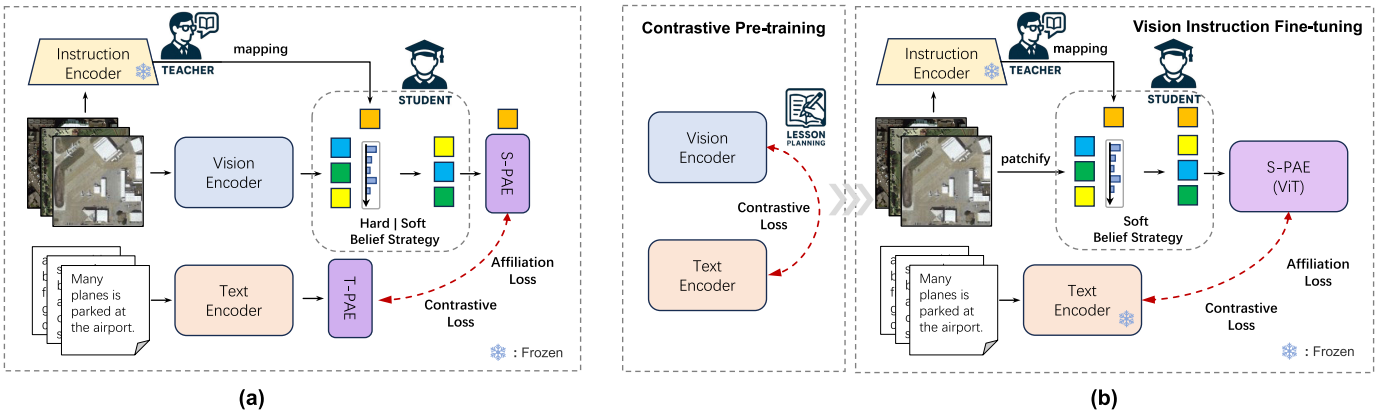


Fig. 2. Both branches of PriorCLIP follow the same visual-prior-guided learning logic while operating at different data scales. The PriorCLIP structure. (a) PriorCLIP for CD-RSITR, which uses S-PAE and T-PAE for unbiased representation in the common subspace (teacher instructs student). (b) PriorCLIP for OD-RSITR, which uses a two-stage strategy to pretrain vision and text encoders on large-scale remote sensing image–text pairs, then perform vision instruction fine-tuning for open-domain retrieval (teacher instructs student through lesson planning).

instruction adaptation, improving coverage of remote sensing concepts and robustness to domain shifts.

4) We introduce a cluster-based symmetric contrastive affiliation loss that incorporates scene-category structure into bidirectional alignment, promoting intraclass compactness and interclass separability to reduce semantic confusion zones.

Compared with our previous work [30] (ACMMM 2023), this article advances RSITR with three points. We first provide a principled synthesis of *closed-domain* versus *open-domain* RSITR, formalizing problem settings and assumptions on label space and domain shift, unifying representative methods, and clarifying evaluation criteria (both closed domain and open domain). Building on this, we extend the visual-prior-guided vision–language model framework to genuinely open-domain retrieval via lightweight, prior-guided adapters that interface with large vision–language backbones, thereby supporting compositional queries and long-tail concepts with minimal task-specific tuning, improved robustness to vocabulary shifts, and efficient inference. Finally, we enhance PriorCLIP by introducing a soft-belief refinement mechanism that assigns continuous belief weights to visual tokens, emphasizing reliable regions while suppressing noisy cues. This effectively mitigates semantic noise and reduces spurious correlations, with benefits particularly pronounced in open-domain scenarios. Extensive experiments across standard RSITR benchmarks confirm consistent gains, with comprehensive ablations and closed-domain and open-domain retrieval studies validating the effectiveness of the proposed PriorCLIP.

II. Related Works

A. Remote Sensing Image–Text Retrieval

Although traditional image–text retrieval methods based on natural images are increasingly mature, they cannot be directly applied to multiscale and redundant remote sensing images. There is still a long way to go in RSITR. According to different training paradigms, RSITR methods can be divided into CD-RSITR and OD-RSITR methods.

Closed-domain method is a supervised method for training and retrieval under a single dataset, but it is less capable of retrieving more complex ground elements. Some works [13], [15], [16], [28], [31] only perform self-interaction on the same modality to obtain an enhanced modal semantic representation. Abdullah et al. [13] and Mao et al. [31] used information of different modalities to obtain a joint semantic representation. Mi et al. [15] proposed a knowledge-based method to solve the problem of coarse-grained textual descriptions. Yuan et al. [16] proposed a network structure that combines global and local information, utilizing attention-based modules to fuse multilevel information dynamically. Zhang et al. [28] designed a reconstruction module to reconstruct the decoupled features, ensuring the maximum retention of information in these features. Some others [14], [24], [32], [33] use information between different attention mechanisms or shared parameter networks for interactive learning. Lv et al. [14] proposed a fusion-based association learning model to fuse image and text information across modalities to improve the semantic relevance of different modalities. Specifically, Yuan et al. [32] employed a shared pattern transfer module to facilitate interaction between modalities, thereby addressing the semantic heterogeneity issue between different modal data. Cheng et al. [33] proposed a semantic alignment module that uses an attention mechanism to enhance the correspondence between image and text. Yuan et al. [24] suggested an asymmetric multimodal feature matching network to adapt multiscale feature inputs while using visual features to guide text representation. These methods are based on partial simulations of the real, complex remote sensing world and fail to learn fine-grained perceptions of the environment. Subsequent studies further improved remote sensing cross-modal retrieval from several perspectives. EnVLF [34] introduced an end-to-end vision–language fusion framework for joint multimodal optimization. SMLGN, VGSGN, and CDMAN enhanced image–text matching through multilevel guidance, visual salient guidance, and multigranularity association, respectively [35], [36], [37]. More recently, PMP-Guard improved retrieval robustness by explicitly exploiting

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pseudo-matched pairs under noisy supervision [38]. Overall, these methods mainly strengthen cross-modal alignment at the feature level, but still remain limited in modeling richer fine-grained semantic understanding in complex remote sensing scenes.

Open-domain method is fine-tuned on a small dataset after pretraining on an additional large-scale dataset, with better ground element recognition. Since the advent of CLIP [20], CLIPs for tasks in remote sensing [21], [22], [23], [39] have emerged. Some work [39] has begun early exploration of fine-tuning CLIP to enhance RSITR, but it has only achieved poor performance. Furthermore, Liu et al. [21] proposed RemoteCLIP, the first RSVLM, to implement language-guided retrieval and zero-shot tasks. Zhang et al. [22] published RS5M, a large-scale remote sensing image–text dataset with more than 5 million images, and utilized geographic information to train the vision–language model GeoRSCLIP. Coincidentally, a new remote sensing benchmark dataset, SkyScript, and a benchmark model, SkyCLIP, were proposed by Wang et al. [23]. However, their goal is mainly to build a foundation for an RSVLM, without further exploring how to enhance the visual representation of remote sensing. Parameter-efficient methods adapt large-scale pretrained vision–language models, such as CLIP or CoOp initialized on large-scale web image–text (WIT) pairs, to RSITR through lightweight tuning strategies, achieving effective domain adaptation with minimal trainable parameters and low computational cost [40], [41], [42]. These approaches depend on the performance of the foundation model, but it may be too burdensome for pretraining. We aim to propose a CLIP-based model that enhances the performance of large-scale, open-domain remote sensing retrieval, building upon the foundation of RSVLM.

Prior-Guided Adaptation of Remote Sensing VLMs: Beyond domain-specific pretraining, recent methods increasingly inject explicit prior knowledge into VLM adaptation. PE-RSITR [40] introduces a lightweight multimodal adapter, CUP [42] uses object-aware prompts and uncertainty estimation, and PERSVL [43] mines historical outputs to improve supervision. For weakly supervised segmentation, ETF [44] expands category semantics with a large language model and fuses them with spatial cues and multigranularity CLIP features. In contrast, PriorCLIP converts a remote sensing scene-recognition prior into instruction-conditioned beliefs over visual tokens, and uses S-PAE, T-PAE, and affiliation loss to refine image–text representations and preserve category structure for both closed- and open-domain bidirectional retrieval.

B. Transformer-Based Cross-Attention Mechanism

Different from self-attention [45], which models dependencies within a single sequence, cross attention is a mechanism in the Transformer architecture that integrates information across two distinct embedding sequences, often originating from different modalities or outputs [46]. Due to its ability to capture cross-modal interactions, cross attention has been widely adopted in various tasks, such as image–text retrieval [9], [47], [48], [49], [50] and recommendation [51], [52], [53].

In image–text retrieval, Lee et al. [47] proposed the *stacked cross-attention* model, which enables contextual alignment between images and sentences. Wei et al. [48] introduced the *multimodality cross-attention* network to jointly capture intramodal and intermodal relationships in a unified framework. Xu et al. [49] developed the CASC framework, which combines cross-modal attention for local alignment with multilabel prediction to enforce global semantic consistency. However, these approaches are often tailored to specific tasks and lack general applicability. To address this, Diao et al. [50] designed two plug-and-play regulators for adaptive contextualization and aggregation of cross-modal features. Liu et al. [9] further proposed a bidirectional correct attention network that adjusts attention weights by modeling the relevance between subfragments and global semantics. Despite their effectiveness, most existing methods struggle to scale with large temporal- and spatial-scale data, highlighting the need for a universal and easily adaptable cross-attention structure.

III. Preliminary: RSITR

A. Generic Image and Text Encoding

A common approach to interpreting remote sensing images [54] is to align visual and textual information at the semantic level. RSITR serves as a fundamental task for this alignment by encoding image and text features into a shared semantic space.

For image representation, existing RSITR methods employ either CNN-based encoders [13], [14], [15], [16], [24], [32], [33], [55] or Transformer-based encoders [56], [57]. Formally, given an RGB image $\mathbf{I}_{\text{img}} \in \mathbb{R}^{3\times H\times W}$, an image encoder $\mathbf{E}_{\text{img}}$ maps it into a global feature $\mathbf{f}_{\text{cls}} \in \mathbb{R}^d$ (corresponding to `[CLS]`) and a set of local features $\mathbf{F}_v \in \mathbb{R}^{d\times m}$

$$[\mathbf{f}_{\text{cls}}, \mathbf{F}_v] = \mathbf{E}_{\text{img}}\left(\mathbf{I}_{\text{img}}; \Theta_{\text{img}}\right) \tag{1}$$

where Θ_{img} denotes the trainable parameters of the encoder, and $[\cdot,\cdot]$ indicates the concatenation along the sequence dimension. For CNN-based encoders, $\mathbf{f}_{\text{cls}}$ captures the high-level semantics, while $\mathbf{F}_v$ corresponds to the spatially aggregated feature maps.

For text representation, RNN-based encoders [13], [16], [24] and Transformer-based encoders [15], [28] are most commonly used. Compared with RNNs [58], Transformers [45] capture global semantic dependencies by attending to all positions in the sequence simultaneously. Given a text sequence $\mathbf{T}_{\text{txt}} = \{w_1, w_2, \ldots, w_n\}$, the text encoder $\mathbf{E}_{\text{txt}}$ yields a global feature $\mathbf{t}_{\text{cls}} \in \mathbb{R}^d$ and local token embeddings $\mathbf{F}_t \in \mathbb{R}^{d\times n}$

$$[\mathbf{t}_{\text{cls}}, \mathbf{F}_t] = \mathbf{E}_{\text{txt}}(\mathbf{T}_{\text{txt}}; \Theta_{\text{txt}}) \tag{2}$$

where Θ_{txt} denotes the trainable parameters of the text encoder, and n is the number of tokens in the input sequence. In RNN-based encoders, $\mathbf{t}_{\text{cls}}$ is often obtained by applying a pooling operation over $\mathbf{F}_t$.

In summary, both $\left[\mathbf{f}_{\text{cls}}, \mathbf{F}_v\right]$ from (1) and $\left[\mathbf{t}_{\text{cls}}, \mathbf{F}_t\right]$ from (2) provide the basis for subsequent cross-modal alignment, while the specific extraction strategies vary depending on encoder architectures.

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B. Various Image–Text Alignment

A straightforward approach to image–text alignment is to fine-tune the vision and text encoders with learnable weights Θ_{img} and Θ_{txt}, aligning the global features $\mathbf{f}_{\text{cls}}$ and $\mathbf{t}_{\text{cls}}$ directly, as in VSE [59]. This method performs well on datasets with abundant paired samples (e.g., COCO [60]), but its improvement is limited in remote sensing scenarios.

To capture richer semantics, several works such as SCAN [55], CAMP [47], and CAMERA [61] introduce feature-level interactions between image and text representations. This can be formalized as

$$\mathbf{Z}_{\text{img-txt}} = \mathbf{E}_{\text{img-txt}}\left(\mathbf{F}_v, \mathbf{F}_t; \Theta_{\text{img-txt}}\right) \tag{3}$$

where $\mathbf{E}_{\text{img-txt}}$ denotes a cross-modal interaction module parameterized by $\Theta_{\text{img-txt}}$. While effective in natural image retrieval, these methods face challenges in remote sensing due to variable imaging scales and the difficulty of fine-grained alignment.

In most RSITR approaches, visual information is further refined since remote sensing images often contain substantial redundancy. For instance, AMFMN [24] and SWAN [17] perform multiscale fusion of $\mathbf{f}_{\text{cls}}$ and $\mathbf{F}_v$, filtering redundant features to highlight salient visual cues, GaLR.

IV. Visual-Prior-Guided Vision–Language Model

A. Instruction Encoding

Remote sensing images often suffer from semantic noise, where background clutter or irrelevant objects bias the visual representation. Recent studies [62], [63], [64], [65], [66], [67] have shown that introducing prompts or instructions can effectively guide models toward more reliable representations. Inspired by this idea, we adopt a ResNet [68] pretrained on the AID dataset [69] as an instruction encoder

$$\mathbf{f}_{\text{ins}} = \mathbf{E}_{\text{ins}}\left(\mathbf{I}_{\text{img}}; \Theta_{\text{ins}}\right) \tag{4}$$

where Θ_{ins} denotes the pretrained weights. The encoder produces an instruction embedding $\mathbf{f}_{\text{ins}} \in \mathbb{R}^d$, which provides prior knowledge for refining visual features.

Unlike the GaLR [16] method, which incorporates $\mathbf{F}_v$ with external object-level knowledge via GCN and fuses them directly—potentially disturbing intrinsic image semantics—our approach treats instruction embeddings as *guidance* rather than as additional features. Specifically, the instruction embedding $\mathbf{f}_{\text{ins}}$ is used to filter and reweight the raw visual features $\left[\mathbf{f}_{\text{cls}}, \mathbf{F}_v\right]$ through an image–instruction interaction module

$$\mathbf{Z}_{\text{img-ins}} = \mathbf{E}_{\text{img-img}}\left(\left[\mathbf{f}_{\text{cls}}, \mathbf{F}_v\right]; \mathbf{f}_{\text{ins}}, \Theta_{\text{img-img}}\right). \tag{5}$$

In this way, the model selectively emphasizes unbiased and semantically reliable visual features, reducing noise and improving the quality of visual representations.

B. Hard and Soft Belief Strategy

Remote sensing images often contain small-scale objects that are easily disturbed by background clutter or irrelevant regions, which amplifies semantic noise in visual representations. To mitigate this issue, we design a belief strategy that leverages instruction embeddings to construct a belief matrix, guiding the filtering of visual features toward an unbiased representation, as illustrated in Fig. 2.

1) Hard Belief Strategy: Given prior knowledge from remote sensing scene recognition $\mathbf{f}_{\text{ins}} \in \mathbb{R}^d$, we compute a belief matrix $\mathbf{M}^{\text{bel}} \in \mathbb{R}^{1\times(m+1)}$ over the concatenated features $\left[\mathbf{f}_{\text{cls}}, \mathbf{F}_v\right]$

$$\mathbf{M}^{\text{bel}} = \text{Softmax}\left(\mathbf{f}_{\text{ins}}^{\text{T}}\left[\mathbf{f}_{\text{cls}}, \mathbf{F}_v\right]\right). \tag{6}$$

Features are then sorted and filtered according to $\mathbf{M}^{\text{bel}}$ to retain the top-k visual tokens

$$\begin{aligned}\left[\mathbf{f}_{\text{cls}}, \mathbf{F}_v\right] &\xrightarrow[\text{Sort\&Filter}]{\mathbf{M}^{\text{bel}}} \mathbf{F}_v^{(k)} \\ \mathbf{F}_v^{(k)} &= \left[\mathbf{F}_{:,r_1}, \ldots, \mathbf{F}_{:,r_k}\right], \quad (r_k < m+1).\end{aligned} \tag{7}$$

This hard filtering enforces discrete feature selection but requires a manually set k, making it sensitive to empirical choices.

2) Soft Belief Strategy: To avoid fixed feature selection, we propose a soft belief strategy that dynamically reweights features instead of discarding them. For each element in $\mathbf{M}^{\text{bel}}$, we compute its batchwise rank R_j

$$R_j = 1 + \sum_{k=1}^{m+1} \mathbb{I}\left(\mathbf{M}_k^{\text{bel}} < \mathbf{M}_j^{\text{bel}}\right) \tag{8}$$

where $\mathbb{I}(\cdot)$ is the indicator function. Each visual token $\mathbf{F}_{:,l} \in \mathbb{R}^d$ is then reweighted based on both its belief score and relative rank

$$\mathbf{F}_v^{(\text{soft})} = \sum_{l=1}^{m+1} \mathbf{F}_{:,l}\left(\mathbf{M}_l^{\text{bel}} + \frac{1}{\sqrt{R_l}}\right). \tag{9}$$

This produces rank-weighted features that preserve all tokens while emphasizing reliable ones. Unlike (7), this strategy requires no predefined k, offering better adaptability. The soft method requires only a single training run and avoids the repeated experiments needed to tune the threshold k in the hard method, thus significantly improving overall efficiency.

C. Visual-Prior-Guided Image–Text Alignment

1) Progressive Attention Encoder Layer: The Transformer encoder layer (TEL) is a fundamental component of the Transformer architecture and has been widely applied in both natural language processing and computer vision tasks [70], [71], [72]. Building upon TEL, we design a progressive attention encoder layer (PAEL) that integrates both self-attention [45] and cross attention [73] to progressively enhance feature representations.

Unlike self-attention, which models dependencies within a single sequence, cross attention enables interaction between two distinct sequences, regardless of modality. Given two inputs $\mathbf{H}_{l-1}^S \in \mathbb{R}^{d\times N}$ and $\mathbf{H}_{l-1}^C \in \mathbb{R}^{d\times N}$, PAEL is computed as

$$\mathbf{H}_{l+1}^S = \text{FFN}\left(\text{Attn}\left(\mathbf{H}_{l-1}^S\right)\right) \tag{10}$$

$$\mathbf{H}_{l+1}^C = \text{FFN}\left(\text{Attn}\left(\mathbf{H}_{l-1}^C, \mathbf{H}_{l+1}^S\right)\right) \tag{11}$$

where Attn(·) denotes multihead attention and FFN(·) represents a feedforward multilayer perceptron.

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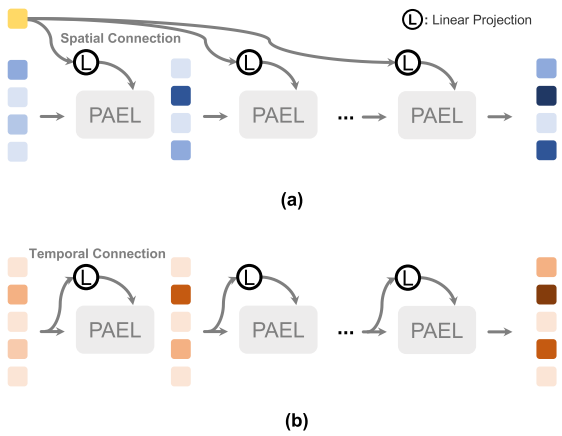


Fig. 3. Two different PAEs for messaging. (a) S-PAE. (b) T-PAE.

In essence, PAEL first strengthens the self-representation of sequence $\mathbf{H}_{l-1}^{S}$ through self-attention [see (10)], and then uses the refined $\mathbf{H}_{l+1}^{S}$ as contextual guidance for updating $\mathbf{H}_{l-1}^{C}$ via cross attention [see (11)]. This progressive mechanism allows information to be propagated step by step, enhancing both intrasequence dependencies and intersequence interactions. For convenience, the PAEL computation can be compactly expressed as

$$PAEL\left(\mathbf{H}_{l-1}^{S}, \mathbf{H}_{l-1}^{C}\right). \tag{12}$$

Compared with parallel attention designs, PAEL adopts a progressive update strategy to ensure that cross attention operates on refined and semantically enriched features. This not only reduces the risk of propagating noisy information across modalities but also facilitates more stable and discriminative feature learning.

2) Progressive Attention Encoder: To effectively capture key feature representations, we design two complementary message transfer mechanisms based on PAEL: S-PAE and T-PAE, as illustrated in Fig. 3.

S-PAE [see Fig. 3(a)] establishes spatial connections between the input sequence and an external source through linear projection, where the interaction is realized by cross attention rather than direct feature fusion [16], [17], [74]. In conventional PAELs, self-attention alone struggles to capture global information, since long-range dependencies require extended sequence distances to be connected. To overcome this limitation, S-PAE leverages external guidance to model long-range dependencies on filtered features activated by prior information

$$\begin{aligned}\mathbf{F}_v^i &= \mathrm{PAEL}\left(\mathbf{F}_v^{i-1}, \mathbf{W}_s^i \mathbf{F}_{\mathrm{ins}}\right)\\ i &= 1, 2, \ldots, n_v, \qquad \mathbf{F}_v^0 = \mathbf{F}_v\end{aligned} \tag{13}$$

where $\mathbf{F}_{\mathrm{ins}} = \left[\mathbf{f}_{\mathrm{ins}}, \mathbf{f}_{\mathrm{ins}}, \ldots, \mathbf{f}_{\mathrm{ins}}\right] \in \mathbb{R}^{d\times r_k}$ denotes the replicated instruction embeddings, $\mathbf{W}_s^i \in \mathbb{R}^{d\times d}$ is the ith projection weight, and we write $\mathbf{F}_v^i = \left[\mathbf{f}_{r_1}^i, \ldots, \mathbf{f}_{r_k}^i\right]$ for the tokenwise output at layer i. From the last PAEL, we obtain an unbiased local-correlated embedding as

$$\mathbf{f}_{\mathrm{loc}} = \mathrm{Head}\left(\mathbf{f}_{r_1}^{n_v}\right) \tag{14}$$

where $\mathrm{Head}(\cdot)$ maps the head token to an unbiased representation. Finally, to preserve global semantic correlations, the overall visual embedding is derived by combining global and local information

$$\mathbf{v}_{\mathrm{emb}} = \mathbf{f}_{\mathrm{cls}} + \mathbf{f}_{\mathrm{loc}}. \tag{15}$$

S-PAE enables global dependency modeling under the guidance of instruction embeddings, producing unbiased and semantically enriched visual representations that complement the standard self-attention pathway.

T-PAE [see Fig. 3(b)] establishes temporal connections across successive time steps through linear projection. While positional encoding is commonly used to capture sequential order, it may introduce ambiguity or inaccuracies when applied to deeper self-attention layers. To address this limitation, T-PAE computes attention maps jointly from the outputs of the previous and current time steps, thereby providing more accurate positional cues and activating informative textual features.

Formally, the recursive update of T-PAE is defined as

$$\begin{aligned}\mathbf{F}_t^i &= \mathrm{PAEL}\left(\mathbf{F}_t^{i-1}, \mathbf{W}_t^i \mathbf{F}_t^{i-1}\right)\\ i &= 1, 2, \ldots, n_t, \qquad \mathbf{F}_t^0 = [\mathbf{t}_{\mathrm{cls}}, \mathbf{F}_t]\end{aligned} \tag{16}$$

where $\mathbf{W}_t^i \in \mathbb{R}^{d\times d}$ is the ith projection weight, and $\mathbf{F}_t^i = \left[\mathbf{t}_{\mathrm{cls}}^i, \mathbf{e}_1^i, \mathbf{e}_2^i, \ldots, \mathbf{e}_n^i\right]$ denotes the tokenwise output at layer i.

From the last PAEL, an unbiased local-correlated embedding is obtained as

$$\mathbf{t}_{\mathrm{loc}} = \mathrm{Head}\left(\mathbf{t}_{\mathrm{cls}}^{n_t}\right) \tag{17}$$

where $\mathrm{Head}(\cdot)$ maps the head token into a compact semantic representation. Finally, to preserve global semantic information, the overall text embedding is derived by combining global and local representations

$$\mathbf{t}_{\mathrm{emb}} = \mathbf{t}_{\mathrm{cls}} + \mathbf{t}_{\mathrm{loc}}. \tag{18}$$

T-PAE complements standard positional encoding by progressively propagating temporal dependencies across layers, enabling more reliable and unbiased text embeddings for retrieval tasks.

3) PriorCLIP for Closed-Domain and Open-Domain Retrieval: To address semantic noise and enhance cross-modal alignment, we extend PriorCLIP with a soft-belief strategy, improving upon the previous hard-belief mechanism [30]. Specifically, we first collect positive remote sensing image–text pairs $\left\{(\mathbf{I}_{\mathrm{img}}^1, \mathbf{T}_{\mathrm{txt}}^1), \ldots, (\mathbf{I}_{\mathrm{img}}^m, \mathbf{T}_{\mathrm{txt}}^m)\right\}$ from dataset $\mathcal{D}_{\mathrm{img\text{-}txt}}$. The image $\mathbf{I}_{\mathrm{img}}$ and text $\mathbf{T}_{\mathrm{txt}}$ are then encoded separately to obtain visual features $\left[\mathbf{f}_{\mathrm{cls}}, \mathbf{F}_v\right]$, instruction embeddings $\mathbf{f}_{\mathrm{ins}}$, and textual features $\left[\mathbf{t}_{\mathrm{cls}}, \mathbf{F}_t\right]$. Next, we compute the belief matrix $\mathbf{M}_j^{\mathrm{bel}}$ between visual features and instruction embeddings, and calculate its rank R_j in batch training. Each element $\mathbf{F}_{:,l}$ of $\left[\mathbf{f}_{\mathrm{cls}}, \mathbf{F}_v\right] \in \mathbb{R}^{d\times(m+1)}$ is then refined via rank-weighted aggregation to obtain $\mathbf{F}_v^{(\mathrm{soft})}$. Finally, S-PAE and T-PAE are applied to capture long-range dependencies, yielding the final visual embedding $\mathbf{v}_{\mathrm{emb}}$ and text embedding $\mathbf{t}_{\mathrm{emb}}$, respectively. This design provides an unbiased vision representation and strengthens textual encoding through cyclic activation.

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TABLE I

COMPARISON OF CLOSED-DOMAIN IMAGE–TEXT RETRIEVAL RESULTS ON RSICD AND RSITMD DATASETS. BEST RESULTS ARE GIVEN IN BOLD, AND SECOND-BEST RESULTS ARE GIVEN IN UNDERLINED

Method	Params	Backbone Vision / Text	RSICD Dataset			RSITMD Dataset		
			Image→Text R@1 / R@5 / R@10	Text→Image R@1 / R@5 / R@10	mR	Image→Text R@1 / R@5 / R@10	Text→Image R@1 / R@5 / R@10	mR
$VSE_{0\,2017}^{TL}$ [59]	29M	ResNet-50 / GRU	4.56 / 16.73 / 22.94	4.37 / 15.37 / 25.35	14.89	9.07 / 21.61 / 31.78	7.73 / 27.80 / 41.00	23.17
$SCAN_{2018}^{i2t}$ [47]	60M	RoI Transformer / GRU	4.82 / 13.66 / 21.99	3.93 / 15.20 / 25.53	14.19	8.92 / 22.12 / 33.78	7.43 / 25.71 / 39.03	22.83
$SCAN_{2018}^{t2i}$ [47]	60M	RoI Transformer / GRU	4.79 / 16.19 / 24.86	3.82 / 15.70 / 28.28	15.61	7.01 / 20.58 / 30.90	7.06 / 26.49 / 42.21	22.37
$CAMP_{2021}$ [55]	63M	RoI Transformer / GRU	4.64 / 14.61 / 24.09	4.25 / 15.82 / 27.82	15.20	8.11 / 23.67 / 34.07	6.24 / 26.37 / 42.37	23.47
$CAMERA_{2020}$ [61]	64M	RoI Transformer / BERT	4.57 / 13.08 / 21.77	4.00 / 15.93 / 26.97	14.39	8.33 / 21.83 / 33.11	7.52 / 26.19 / 40.72	22.95
$LW\text{-}MCR_{2021}$ [55]	–	CNNs / CNNs	3.29 / 12.52 / 19.93	4.66 / 17.51 / 30.02	14.66	10.18 / 28.98 / 39.82	7.79 / 30.18 / 49.78	27.79
$AMFMN_{2022}$ [24]	36M	ResNet-18 / GRU	5.21 / 14.72 / 21.57	4.08 / 17.00 / 30.60	15.53	10.63 / 24.78 / 41.81	11.51 / 34.69 / 54.87	29.72
$GaLR_{2022}$ [16]	46M	ResNet-18 + PP-YOLO / GRU	6.59 / 19.85 / 31.04	4.69 / 19.48 / 32.13	18.96	14.82 / 31.64 / 42.48	11.15 / 36.68 / 51.68	31.41
KCR_{2022} [15]	–	ResNet-101 / BERT	5.95 / 18.59 / 29.58	5.40 / 22.44 / 37.36	19.89	–	–	–
$SWAN_{2023}$ [17]	40M	ResNet-50 / GRU	7.41 / 20.13 / 30.86	5.56 / 22.26 / 37.41	20.61	13.35 / 32.15 / 46.90	11.24 / 40.40 / 60.60	34.11
$HVSA_{2023}$ [75]	–	ResNet-50 / GRU	7.47 / 20.62 / 32.11	5.51 / 21.13 / 34.13	20.61	13.20 / 32.08 / 45.58	11.43 / 39.20 / 57.45	33.16
$DOVE_{2024}$ [74]	38M	ResNet-50 + RoI Transformer / GRU	8.66 / 22.35 / 34.95	6.04 / 23.95 / 40.35	22.72	16.81 / 36.80 / 49.93	12.20 / **49.93** / **66.50**	37.73
$VSE_{1\,2017}^{CL}$ [59]	131M	ViT-B / BERT	9.06 / 22.78 / 32.75	5.32 / 19.47 / 33.71	20.52	12.83 / 31.19 / 46.24	9.60 / 36.59 / 54.42	31.81
$VSE_{2\,2017}^{TL}$ [59]	137M	Swin-T / BERT	7.96 / 21.68 / 34.31	6.44 / 22.09 / 37.49	21.66	17.70 / 36.28 / 48.67	13.58 / 41.24 / 59.29	36.13
$VSE_{2\,2017}^{CL}$ [59]	137M	Swin-T / BERT	10.52 / 25.07 / 36.78	6.11 / 23.64 / 38.28	23.40	16.15 / 40.04 / 53.10	10.75 / 38.98 / 60.18	36.53
$VSE_{2\,2017}^{CL}$ [59]	161M	Swin-T / BERT	10.43 / 25.89 / 37.15	6.02 / 23.42 / 38.30	23.53	17.48 / 36.95 / 50.44	11.81 / 41.99 / 61.77	36.74
$EnVLF_{2023}$ [34]	-	Vision Transformer / BERT	7.78 / 20.52 / 31.56	6.09 / 22.33 / 36.65	20.82	13.42 / 33.11 / 47.12	11.15 / 38.63 / 57.58	33.50
$SMLGN_{2024}$ [35]	-	Vision Transformer / BERT	8.87 / 25.53 / 37.24	7.85 / 27.14 / 42.58	24.87	17.26 / 39.38 / 51.55	13.19 / 43.94 / 60.40	37.62
$VGSGN_{2024}$ [36]	-	Pyramid Vision Transformer / BERT	8.33 / 21.87 / 32.57	6.53 / 23.13 / 36.85	21.55	14.16 / 34.96 / 50.66	13.23 / 42.57 / 63.41	36.50
$CDMAN_{2024}$ [37]	-	Pyramid Vision Transformer / BERT	10.23 / 24.94 / 36.83	7.13 / 25.38 / 40.35	24.14	16.46 / 37.74 / 51.24	**13.82** / 41.96 / 59.74	36.83
$PMPGuard_{2026}$ [38]	-	Swin-T / BERT	12.56 / 28.20 / 39.37	9.76 / 25.55 / 42.49	26.21	14.94 / 37.39 / 49.11	12.58 / 37.06 / 59.01	34.95
PriorCLIP-CD (hard)	161M	Swin-T + ResNet-50 / BERT	9.88 / **27.26** / **39.16**	6.97 / 24.56 / 38.92	24.46	18.14 / 41.15 / 52.88	12.17 / 41.68 / 63.41	38.24
PriorCLIP-CD (soft)	161M	Swin-T + ResNet-50 / BERT	**10.89** / 26.17 / 37.79	**7.17** / **25.07** / **41.06**	**24.69**	**18.36** / **42.04** / **55.53**	13.36 / 44.47 / 61.73	**39.25**

For open-domain retrieval, we propose a two-stage prior representation learning strategy. In the first stage, the model is pretrained on the large-scale RS5M dataset [22] with contrastive loss (CL) to learn generalizable vision–language representations. In the second stage, fine-tuning is performed on RSITR datasets, following a procedure similar to the closed-domain case but with two key distinctions: 1) the visual backbone is updated based on base ViT and ResNet models and 2) the text encoder is replaced with a language Transformer trained on a large-scale remote sensing corpus, eliminating the need for an additional S-PAE, which is based on a simplified variant of our visual-prior-guided vision–language model. Moreover, PriorCLIP further strengthens the ViT architecture by incorporating convolutional patching, a soft-belief refinement strategy, and a Transformer encoder to enhance interactions between visual and instructional features.

In summary, PriorCLIP introduces: 1) a soft-belief strategy to mitigate semantic noise; 2) S-PAE and T-PAE for robust closed-domain retrieval; and 3) a two-stage prior learning scheme with architecture enhancements for open-domain retrieval, collectively leading to improved adaptability and retrieval performance across both settings. From a unified perspective, both regimes follow the same high-level sequence: a remote sensing prior is extracted, the prior guides representation adaptation, and the resulting image and text embeddings are optimized in a common semantic space. Closed-domain retrieval applies this sequence within paired samples through the belief strategy and PAE modules, while open-domain retrieval extends it across pretraining and fine-tuning stages. The components are therefore complementary realizations of one PriorCLIP framework rather than independent modules.

D. Loss Function

1) Contrastive Loss: In RSITR, TL [12] is commonly used for model optimization. However, CL provides better optimization performance, as shown in the comparison between VSE_2 (TL) and VSE_2 (CL) in Table I. Therefore, we adopt CL [76] to optimize our model, as it requires only pairwise similarity calculations without the need for additional negative samples.

Given the $\mathbf{v}_{emb}$ and $\mathbf{t}_{emb}$ embedded by PriorCLIP from batch image–text pairs, we calculate the cosine similarity between them as $s_{i,j} = \mathbf{v}_i^{\mathrm{T}}\mathbf{t}_j$. Then, we can get the CL as

$$\mathcal{L}_c = -\frac{1}{N}\sum_{i=1}^{N}\left[\log\frac{\exp\left(s_{i,i}/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_{i,j}/\tau\right)} + \log\frac{\exp\left(s_{i,i}/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_{j,i}/\tau\right)}\right] \tag{19}$$

where τ is the temperature parameter.

2) Affiliation Loss: Most existing retrieval tasks rely primarily on pairwise objectives such as CL [20], [77] or TL [12]. While effective, these objectives treat samples independently and overlook the underlying distributional structure of the data [see Fig. 4(a)]. This often results in semantic confusion zones, where samples from different categories are not well separated.

To address this limitation, we propose an affiliation loss, which explicitly exploits the distributional characteristics of remote sensing data by minimizing the distance between each modality and the cluster center of its affiliated category [17], as illustrated in Fig. 4(b).

Affiliation loss is a cluster-based objective [78] that integrates contrastive learning with scene category supervision to

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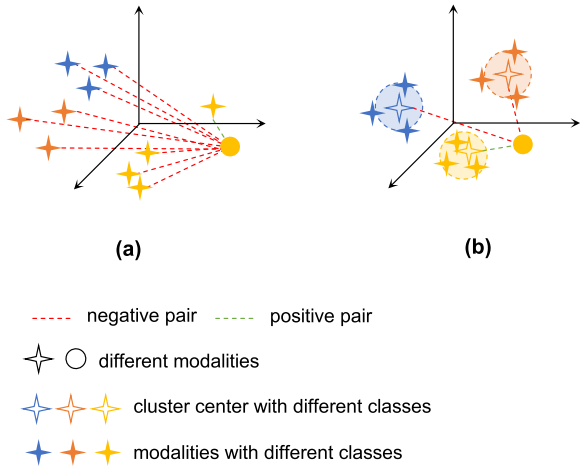


Fig. 4. (a) Pairwise loss. (b) Clusterwise loss.

Algorithm 1 Cluster-Based Symmetric Contrastive Affiliation Loss in a Mini-Batch

Input: Image embeddings $\mathbf{I}_e \in \mathbb{R}^{B\times d_e}$,
Text embeddings $\mathbf{T}_e \in \mathbb{R}^{B\times d_e}$,
Labels $\mathbf{c} = (c_1, \ldots, c_B)^\top \in \{1, \ldots, C\}^B$,
Temperature $t \in \mathbb{R}$,
Small constant $\varepsilon > 0$
Output: Symmetric contrastive loss $\mathcal{L}$

begin

// Step 1: ℓ_2 normalization
$\tilde{\mathbf{I}}_e \leftarrow \mathrm{diag}\big(\|\mathbf{I}_{e,i}\|_2^{-1}\big)\mathbf{I}_e$;
$\tilde{\mathbf{T}}_e \leftarrow \mathrm{diag}\big(\|\mathbf{T}_{e,i}\|_2^{-1}\big)\mathbf{T}_e$;
// Step 2: One-hot label matrix and class counts
Construct $\mathbf{Y} \in \{0,1\}^{B\times C}$ with $Y_{ik} = \mathbb{1}\{c_i = k\}$;
$\mathbf{s} \leftarrow \mathbf{Y}^\top \mathbf{1}_B + \varepsilon \mathbf{1}_C$;
// Step 3: Class prototypes in the batch
$\mathbf{I}_{\mathrm{cls}} \leftarrow \mathrm{diag}(\mathbf{s})^{-1}\mathbf{Y}^\top \tilde{\mathbf{I}}_e$;
$\mathbf{T}_{\mathrm{cls}} \leftarrow \mathrm{diag}(\mathbf{s})^{-1}\mathbf{Y}^\top \tilde{\mathbf{T}}_e$;
// Step 4: Sample-aligned centers
$\mathbf{I}_c \leftarrow \mathbf{Y}\mathbf{I}_{\mathrm{cls}}$;
$\mathbf{T}_c \leftarrow \mathbf{Y}\mathbf{T}_{\mathrm{cls}}$;
// Step 5: Scaled pairwise cosine similarity
$\mathbf{Z}_{I2T} \leftarrow e^t \tilde{\mathbf{I}}_e \mathbf{T}_c^\top$;
$\mathbf{Z}_{T2I} \leftarrow e^t \tilde{\mathbf{T}}_e \mathbf{I}_c^\top$;
// Step 6: Symmetric loss
Let $\mathbf{y} = (0, 1, \ldots, B-1)^\top$;
$\mathcal{L}_{I2T} \leftarrow \mathrm{CE}(\mathbf{Z}_{I2T}, \mathbf{y})$;
$\mathcal{L}_{T2I} \leftarrow \mathrm{CE}(\mathbf{Z}_{T2I}, \mathbf{y})$;
$\mathcal{L} \leftarrow \frac{1}{2}\big(\mathcal{L}_{I2T} + \mathcal{L}_{T2I}\big)$;
return $\mathcal{L}$;

enforce tighter intraclass compactness and clearer interclass separability. As shown in Algorithm 1, samples in a mini-batch are first grouped into C categories based on scene-level annotations.[1] Each image and its paired text are assigned to the same category, yielding positive pairs $\{(\mathbf{v}_i, \mathbf{t}_i^*) \mid i = 1, 2, \ldots, N\}$, where $\mathbf{t}_i^*$ denotes the cluster center of the ith text. Symmetrically, we define vision-centered clusters for text samples.

The vision-to-text and text-to-vision cosine similarities are computed as $s_{i,j}^v = \mathbf{v}_i^\mathrm{T}\mathbf{t}_j^*$ and $s_{j,i}^t = \mathbf{t}_j^\mathrm{T}\mathbf{v}_i^*$, respectively. The cluster-based symmetric CL is then formulated as

$$\mathcal{L}_a = -\frac{1}{N}\sum_{i=1}^{N}\left[\log\frac{\exp\left(s_{i,i}^v/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_{i,j}^v/\tau\right)} + \log\frac{\exp\left(s_{i,i}^t/\tau\right)}{\sum_{j=1}^{N}\exp\left(s_{j,i}^t/\tau\right)}\right] \tag{20}$$

where τ is a temperature scaling factor. The final optimization objective combines the conventional CL $\mathcal{L}_c$ with our proposed affiliation loss

$$\mathcal{L}_{total} = \mathcal{L}_c + \lambda_{cs}\mathcal{L}_a \tag{21}$$

where λ_{cs} is a balancing coefficient controlling the influence of category clustering. Affiliation loss goes beyond pairwise alignment by leveraging class-level structure, effectively reducing semantic confusion zones and yielding more discriminative representations for remote sensing retrieval.

V. Experiments

A. Datasets Settings

1) RSICD Dataset: The RSICD dataset comprises 10 921 remote sensing images with a resolution of 224 × 224, each paired with five corresponding textual captions. Following the protocol in [16] and [24], we split the dataset into 7862 training images, 1966 validation images, and 1093 test images.

2) RSITMD Dataset: The RSITMD dataset contains 4743 images with a resolution of 256 × 256, each annotated with five captions that are more fine-grained than those in RSICD. Following [16], we adopt the official split of 3435 training images, 856 validation images, and 452 test images.

3) RS5M Dataset: The RS5M dataset [22] comprises approximately 5 million remote sensing image–text pairs, collected by filtering publicly available resources and automatically captioning label-only datasets using pretrained image captioning models. In this work, RS5M serves as the large-scale pretraining corpus for PriorCLIP.

B. Evaluation Metrics

In line with prior RSITR studies [24], [55], we adopt R@K ($K = 1, 5, 10$) and mean recall (mR) as evaluation metrics. R@K measures the percentage of correctly retrieved matches within the top-K results, while mR computes the average of all R@K values, offering a comprehensive assessment of retrieval performance.

C. Implementation Details

All experiments are performed using four NVIDIA A100 GPUs. The specific parameters for closed-domain retrieval and open-domain retrieval are detailed as follows.

For closed-domain retrieval, we adopt the Swin Transformer (Swin-T, tiny version) [79] as the vision encoder, pretrained on

[1]For RSICD and RSITMD, category information can be obtained directly from the image file names.

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ImageNet, and the base version (BERT) as the text encoder, utilizing the official pretrained parameters. Both vision and text features are projected from 768 to 512 dimensions through a linear layer. Self-attention and cross-attention modules are configured with eight heads and a dropout rate of 0.2, while the number of stacked units is set to 2 for S-PAE and 3 for T-PAE. For instruction encoding, we employ ResNet-50, pretrained on the AID dataset [69], using the second-to-last layer (dimension 1024) as the output, which is then linearly mapped to 512 dimensions. The CL and affiliation loss are optimized with a temperature coefficient of 0.07 and a center factor of 1, respectively.

For open-domain retrieval, we employ ResNet-50 + ViT-B as the visual backbone and a Transformer as the textual backbone, aligning with the vanilla CLIP approach. A modified ResNet-50 is pretrained on AID [69], and parameters before the attention pooling layer are frozen during CLIP tuning. Training is conducted for 20 epochs in pretraining (with a batch size of 256) and 10 epochs in fine-tuning (with a batch size of 512), each comprising 1 000 000 training steps per epoch. The source code and pretrained model checkpoints are publicly available.[2]

D. Performance Comparisons

1) State-of-the-Art Methods: We compare our method with both closed-domain and open-domain retrieval approaches.

CD-RSITR: These baselines are grouped as follows.

1) *Generic Image–Text Retrieval Models:* VSE_0 [59], SCAN [47], CAMP [55], and CAMERA [61].
2) *Remote Sensing-Specific Models:* LW-MCR [55], AMFMN [24], GaLR [16], KCR [15], SWAN [17], HVSA [75], and DOVE [74].
3) *Transformer-Based Variants:* VSE_1 [59], VSE_2 [59], VSE_2 [59] + TEL, EnVLF [34], SMLGN [35], VGSGN [36], CDMAN [37] and PMPGuard [38].

For SCAN, CAMP, and CAMERA, we follow prior work and adopt RoI Trans [80] with ResNet-50 pretrained on DOTA [81] to detect salient objects. For remote sensing-specific models, we report the best results from the original papers, as they are difficult to reproduce on RSICD and RSITMD. To ensure fairness, we further extend VSE baselines with Transformer backbones, where VSE_i $(i = 0, 1, 2)$ denotes different encoder configurations, and TEL indicates an additional TEL. Loss functions are annotated as TL or CL. Finally, we include two closed-domain variants of our method: "PriorCLIP-CD (hard)" and "PriorCLIP-CD (soft)," which apply hard- and soft-belief strategies, respectively.

OD-RSITR: These approaches train a visual-semantic embedding model with a dual-tower structure on large-scale image–text datasets using natural language supervision. Representative models include the following.

1) *CLIP* [20], trained on 400M WIT pairs denoted as WIT (CLIP).
2) *RemoteCLIP* [21], pretrained on a mixture of remote sensing datasets, including RET-3 (three retrieval datasets: RSICD, RSITMD, and UCM-Captions [82]), DET-10 (ten detection datasets), and SEG-4 (four segmentation datasets).
3) *GeoRSCLIP* [22], trained on RS5M, a large-scale remote sensing image–text corpus.
4) *SkyCLIP* [23], trained on the SkyScript dataset.
5) *Parameter-efficient method* trained on WIT (CLIP) or WIT(CoOP) like: PE-RSITR [40], CLIP-Ada [41], and CUP [42].

For our method in open-domain retrieval, we pretrain PriorCLIP-OD on RS5M [22], which, to the best of our knowledge, is the largest publicly available remote sensing image–text dataset. For fairness, the results of CLIP and SkyCLIP are obtained using officially released weights, whereas the results of RemoteCLIP and GeoRSCLIP are reported from their original papers. RET-3 is constructed by combining three retrieval datasets and further deduplicated following RemoteCLIP [21] to prevent data leakage. In addition, we construct RET-2 (RSICD + RSITMD), which is also deduplicated, as an alternative fine-tuning set for GeoRSCLIP.

2) Quantitative Comparison of Closed-Domain Methods: Closed-domain experiments are conducted under the setting where training and testing are performed on the same dataset, as summarized in Table I.

Results on RSICD: As presented in Table I, remote sensing-specific approaches consistently outperform generic image–text retrieval baselines. Within Transformer-based encoders, the Swin Transformer [83] provides more robust visual features than the ViT [26]. A comparison between VSE_2 trained with TL and CL further indicates that CL is more effective. Retrieval performance also varies significantly depending on the backbone configuration: models with Swin-T combined with BERT perform best, followed by ViT with BERT, while ResNet-50 combined with GRU yields the weakest results. This demonstrates the advantage of Transformer-based architectures over CNN-based ones for RSITR tasks.

The addition of a TEL enables VSE_2+ TEL to achieve an mR of 23.53, although the improvement over VSE_2 remains limited. In contrast, our proposed PriorCLIP substantially enhances retrieval performance, achieving an mR of 24.69, which represents a 4.9% relative improvement over the strongest baseline.

Results on RSITMD: Compared with RSICD, the RSITMD dataset contains more fine-grained textual descriptions, resulting in overall higher retrieval performance. As shown in Table I, traditional image–text retrieval methods remain unsatisfactory across both datasets. Among remote sensing-specific models, SWAN, which combines CNN with GRU, achieves an mR of 34.11. Transformer-based methods proffer further improvements, with $VSE_2 + TEL$ achievingmR of 36.74.

Our PriorCLIP delivers the best performance, attaining an mR of 39.24, which corresponds to a 4.0% relative improvement over the strongest baseline. Moreover, PriorCLIP achieves R@1 scores of 18.36% for image-to-text (I2T) and 13.36% for text-to-image (T2I).

3) Quantitative Comparison of Open-Domain Methods: Open-domain experiments involve fine-tuning on small-scale

[2]https://github.com/jaychempan/PriorCLIP

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TABLE II

COMPARISON OF OPEN-DOMAIN IMAGE–TEXT RETRIEVAL RESULTS ON RSICD AND RSITMD DATASETS. † DENOTES THE ZERO-SHOT CLIP, AND ‡ DENOTES THE FINE-TUNED CLIP. BEST RESULTS ARE GIVEN IN BOLD, AND SECOND-BEST RESULTS ARE GIVEN IN UNDERLINED

Method	Params	Backbone vision / text	Pre-train	Fine-tune	Image→Text R@1 / R@5 / R@10	Text→Image R@1 / R@5 / R@10	mR
RSICD Dataset							
$CLIP_{2021}$ † [20]	151M	ViT-B / Transformer	WIT (CLIP)	-	4.76 / 12.81 / 19.12	4.70 / 15.43 / 25.01	13.64
$CLIP\text{-}FT_{2021}$ ‡ [20]	151M	ViT-B / Transformer	WIT (CLIP)	RSICD	15.55 / 30.56 / 41.99	14.68 / 33.63 / 42.71	29.85
$PE\text{-}RSITR_{2023}$ ‡ [40]	151M	ViT-B / Transformer	WIT (CLIP)	RSICD	14.13 / 31.51 / 44.78	11.63 / 33.92 / 50.73	31.12
$CLIP\text{-}Ada_{2024}$ [41]	-	ViT-B / Transformer	WIT(CoOp)	RSICD	7.67 / 24.87 / 39.73	7.11 / 19.48 / 31.01	21.65
CUP_{2024} [42]	157M	ViT-B / Transformer	WIT (CLIP)	RSICD	12.05 / 32.66 / 48.70	9.86 / 31.60 / 47.15	30.34
$RemoteCLIP_{2024}$ † [21]	151M	ViT-B / Transformer	RET-3+DET-10+SEG-4	-	17.02 / 37.97 / 51.51	13.71 / 37.11 / 54.25	35.26
$RemoteCLIP_{2024}$ † [21]	428M	ViT-L / Transformer	RET-3+DET-10+SEG-4	-	18.39 / 37.42 / 51.05	14.73 / 39.93 / 56.58	36.35
$SkyCLIP_{2024}$ † [23]	151M	ViT-B / Transformer	Skyscript	-	6.59 / 16.10 / 26.53	7.14 / 22.34 / 34.29	18.83
$SkyCLIP_{2024}$ † [23]	428M	ViT-L / Transformer	Skyscript	-	7.04 / 17.75 / 27.08	6.40 / 21.21 / 34.00	18.91
$GeoRSCLIP_{2024}$ † [22]	151M	ViT-B / Transformer	RS5M	-	11.53 / 28.55 / 39.16	9.52 / 27.37 / 40.99	26.18
$GeoRSCLIP\text{-}FT_{2024}$ ‡ [22]	151M	ViT-B / Transformer	RS5M	RSICD	22.14 / 40.53 / 51.78	15.26 / 40.46 / 57.79	38.00
$GeoRSCLIP\text{-}FT_{2024}$ ‡ [22]	151M	ViT-B / Transformer	RS5M	RET-2	21.13 / 41.72 / **55.63**	15.59 / 41.19 / **57.99**	38.87
PriorCLIP-OD ‡	189M	ResNet-50+ViT-B / Transformer	RS5M	RSICD	26.62 / 40.71 / 50.32	**21.35** / 43.59 / 54.27	39.48
PriorCLIP-OD ‡	189M	ResNet-50+ViT-B / Transformer	RS5M	RET-2	25.62 / 43.18 / 53.06	19.91 / 42.74 / 55.15	39.95
PriorCLIP-OD ‡	189M	ResNet-50+ViT-B / Transformer	RS5M	RET-3	**27.63** / **45.38** / 55.26	21.10 / **44.87** / 56.12	**41.73**
RSITMD Dataset							
$CLIP_{2021}$ † [20]	151M	ViT-B / Transformer	WIT (CLIP)	-	5.31 / 17.26 / 25.00	5.84 / 22.74 / 34.82	18.50
$CLIP\text{-}FT_{2021}$ ‡ [20]	151M	ViT-B / Transformer	WIT (CLIP)	RSITMD	25.00 / 48.23 / 62.17	23.72 / 49.38 / 63.63	45.35
$PE\text{-}RSITR_{2023}$ ‡ [40]	151M	ViT-B / Transformer	WIT (CLIP)	RSITMD	23.67 / 44.07 / 60.36	20.10 / 50.63 / 67.97	44.47
$CLIP\text{-}Ada_{2023}$ [41]	-	ViT-B / Transformer	WIT(CoOp)	RSITMD	13.30 / 40.20 / 60.06	12.83 / 28.84 / 39.05	32.38
CUP_{2024} [42]	157M	ViT-B / Transformer	WIT (CLIP)	RSITMD	21.57 / 43.70 / 56.75	17.70 / 48.40 / 64.82	42.16
$RemoteCLIP_{2024}$ † [21]	151M	ViT-B / Transformer	RET-3+DET-10+SEG-4	-	27.88 / 50.66 / 65.71	22.17 / 56.46 / 73.41	49.38
$RemoteCLIP_{2024}$ † [21]	428M	ViT-L / Transformer	RET-3+DET-10+SEG-4	-	28.76 / 52.43 / 63.94	23.76 / 59.51 / 74.73	50.52
$SkyCLIP_{2024}$ † [23]	151M	ViT-B / Transformer	Skyscript	-	10.18 / 25.44 / 35.62	10.88 / 33.27 / 49.82	27.54
$SkyCLIP_{2024}$ † [23]	428M	ViT-L / Transformer	Skyscript	-	12.61 / 28.76 / 38.05	10.62 / 34.73 / 51.42	29.37
$GeoRSCLIP_{2024}$ † [22]	151M	ViT-B / Transformer	RS5M	-	19.03 / 34.51 / 46.46	14.16 / 42.39 / 57.52	35.68
$GeoRSCLIP\text{-}FT_{2024}$ ‡ [22]	151M	ViT-B / Transformer	RS5M	RSITMD	30.09 / 51.55 / 63.27	23.54 / 57.52 / 74.60	50.10
$GeoRSCLIP\text{-}FT_{2024}$ ‡ [22]	151M	ViT-B / Transformer	RS5M	RET-2	32.30 / 53.32 / 67.92	25.04 / **57.88** / 74.38	51.81
PriorCLIP-OD ‡	189M	ResNet-50+ViT-B / Transformer	RS5M	RSITMD	30.97 / 58.19 / 71.02	25.62 / 55.27 / **75.00**	52.68
PriorCLIP-OD ‡	189M	ResNet-50+ViT-B / Transformer	RS5M	RET-2	44.25 / **65.71** / **75.22**	28.45 / 54.65 / 68.05	56.05
PriorCLIP-OD ‡	189M	ResNet-50+ViT-B / Transformer	RS5M	RET-3	**45.58** / 65.49 / 75.00	**30.13** / 55.44 / 68.54	**56.70**

RSITR datasets after pretraining on additional large-scale corpora, as summarized in Table II.

Results on RSICD: The zero-shot performance of CLIP varies considerably depending on the pretraining corpus. For example, RemoteCLIP achieves an mR of 36.35 on RSICD. Results from RemoteCLIP and SkyCLIP further indicate that scaling the backbone from ViT-B to ViT-L yields only marginal gains. Fine-tuning the WIT-trained CLIP model improves mR from 13.64 to 29.85 on RSICD, and further fine-tuning on RSITR datasets yields additional improvements. GeoRSCLIP, fine-tuned on RET-2, achieves an mR of 38.87.

Our PriorCLIP, fine-tuned on RSICD, RET-2, and RET-3, achieves additional improvements of 1.5%, 2.7%, and 7.3%, respectively, over the GeoRSCLIP baseline. Notably, PriorCLIP also delivers the strongest R@1 gains, with 27.63% for I2T and 21.35% for T2I. These results demonstrate the effectiveness of our two-stage CLIP-based framework in enhancing both image and text retrieval.

Results on RSITMD: On RSITMD, RemoteCLIP achieves an mR of 50.52 in the zero-shot setting. Fine-tuning WIT-based CLIP substantially improves performance, raising mR from 18.50 to 45.35. Compared with RemoteCLIP (ViT-L) pretrained on RET-3, PriorCLIP achieves a 9.6% higher mR despite using fewer parameters. GeoRSCLIP fine-tuned on RET-2 further increases mR performance to 51.81. Table V shows that PriorCLIP increases the parameter size by 25.2% (151M → 189M), but its inference time grows by only 15.1%–17.6%, while GFLOPs increase by merely 1.2%. This shows that the added parameters bring only limited runtime overhead. Notably, PriorCLIP attains the lowest time-per-parameter ratio (0.0100 s/M) among the compared models, indicating that its extra parameters are computationally affordable and the resulting inference-time increase is acceptable in practice.

Our PriorCLIP surpasses this benchmark, yielding additional mR gains of 1.7%, 8.2%, and 9.4% when fine-tuned on RSITMD, RET-2, and RET-3, respectively. In terms of R@1, PriorCLIP achieves 45.58% for I2T and 30.13% for T2I. These results highlight the benefits of our vision-instruction fine-tuning strategy in strengthening visual representations and improving open-domain retrieval performance.

4) Visualization of Closed-Domain and Open-Domain Methods: Fig. 5 (PriorCLIP-CD versus PriorCLIP-OD) and Fig. 6 (PriorCLIP versus GeoRSCLIP versus RemoteCLIP) are qualitative examples of the top-5 retrieval results for both `Image-query-Text` (left) and `Text-query-Image` (right). In `Image-query-Text`, the blue box denotes the query image, green text indicates correctly matched captions, and red text corresponds to mismatched results. In `Text-query-Image`, the blue text denotes the query sentence, while green and red boxes represent correct and incorrect image matches, respectively.

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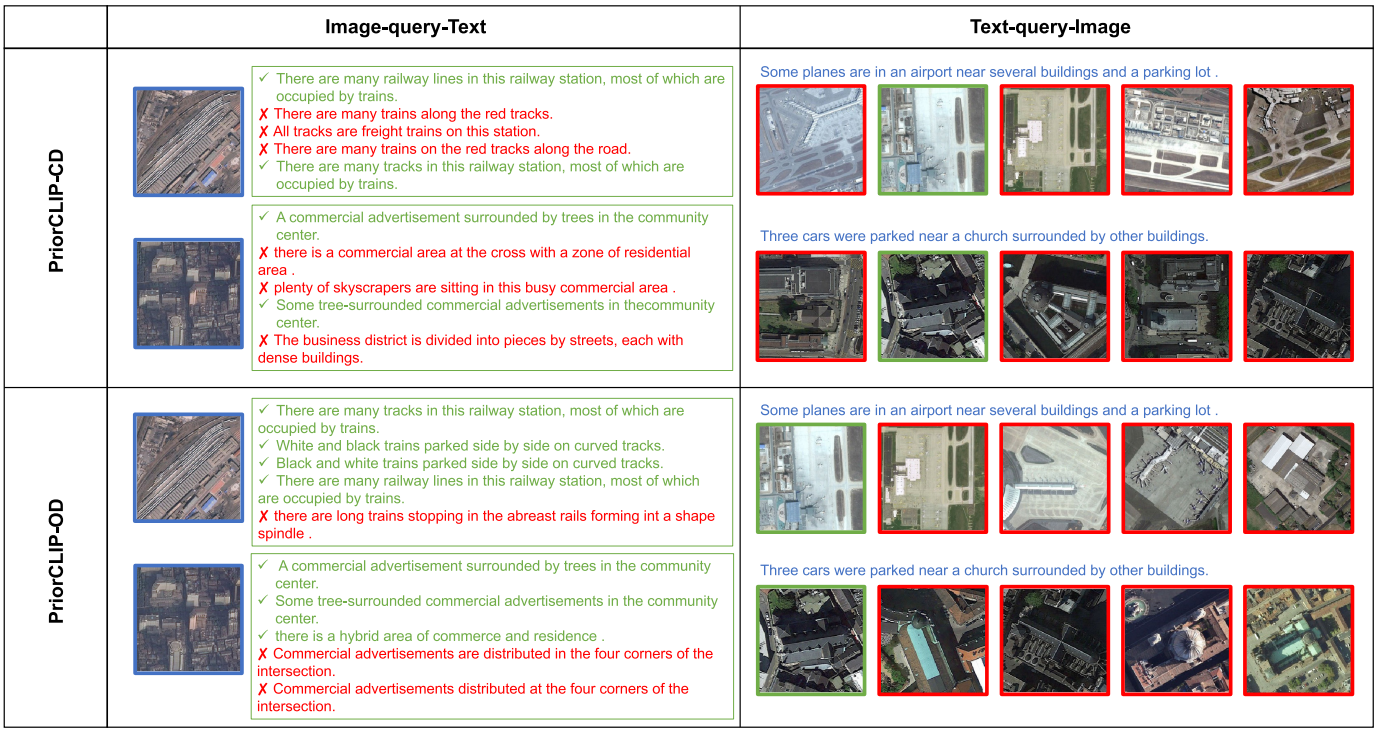


Fig. 5. Visualization of PirorCLIP-CD and PirorCLIP-OD on the top 5 retrieval results includes (left) image–query–text and (right) text–query–image. Blue text and boxes represent the queries, green text and boxes represent the matching retrieval results, and red text and boxes indicate the mismatched retrieval results.

For `Image-query-Text`, most of the top-5 retrieved captions correctly correspond to the query image, despite each image being associated with only five ground-truth captions. For `Text-query-Image`, the correct image is typically ranked first or second among the retrieved results, even though each caption is paired with only a single ground-truth image. We also mark the correct and incorrect matches to more clearly demonstrate the advantage of PriorCLIP in reducing semantic confusion and improving retrieval accuracy compared with GeoRSCLIP and RemoteCLIP in Fig. 6.

Overall, PriorCLIP provides more semantically coherent retrievals and assigns higher ranks to correct matches compared with competing methods. In `Text-query-Image`, although some retrieved images are visually similar but semantically less precise, the ranking gap between correct and incorrect results is larger under PriorCLIP, suggesting that our model effectively suppresses misleading candidates. Nonetheless, certain failure cases remain, where semantically related but incorrect results are retrieved, highlighting opportunities for further refinement.

TABLE III

ABLATION ON THE RSITMD TEST SET. † DENOTES A MODIFIED S-PAE. BEST RESULTS ARE GIVEN IN BOLD, AND SECOND-BEST RESULTS ARE GIVEN IN UNDERLINED

Method	Image→Text R@1 / R@5 / R@10	Text→Image R@1 / R@5 / R@10	mR
PriorCLIP for Closed-Domain RSITR			
baseline	16.15 / 40.04 / 53.10	10.75 / 38.98 / 60.18	36.53
+*S-PAE*	16.59 / 38.50 / 54.65	12.88 / 40.04 / 58.19	36.81
+*T-PAE*	**20.13** / 41.15 / 53.10	**14.60** / 38.89 / 53.81	36.95
+$\mathcal{L}_a$	19.25 / 36.95 / 52.88	10.88 / 39.51 / 59.87	36.56
+*S-PAE*+*T-PAE*	17.04 / 39.16 / 54.42	13.50 / 40.04 / 59.34	37.25
+*S-PAE*+$\mathcal{L}_a$	16.81 / 40.04 / **55.31**	11.59 / 41.50 / 61.02	37.71
+*T-PAE*+$\mathcal{L}_a$	17.04 / 38.72 / 53.76	11.81 / 41.64 / 60.58	37.26
+*S-PAE*+*T-PAE*+$\mathcal{L}_a$	18.14 / **41.15** / 52.88	12.17 / **41.68** / **63.41**	**38.24**
PriorCLIP for Open-Domain RSITR			
CLIP-baseline	25.00 / 48.23 / 62.17	23.72 / 49.38 / 63.63	45.35
+*S-PAE* †	24.56 / 53.32 / 66.37	21.02 / 51.86 / 71.02	48.02
+$\mathcal{L}_a$	27.65 / 52.21 / 63.94	25.49 / 53.10 / 66.59	48.16
+*S-PAE* † + $\mathcal{L}_a$	**30.97** / **58.19** / **71.02**	**25.62** / **55.27** / **75.00**	**52.68**

E. Ablation Studies

We conduct 12 ablation experiments (eight for PriorCLIP-CD and four for PriorCLIP-OD) to evaluate the contribution of three key components: S-PAE, T-PAE, and the proposed affiliation loss $\mathcal{L}_a$, as summarized in Table III. The *baseline* setting removes all three modules, while *CLIP-baseline* denotes PriorCLIP without S-PAE and affiliation loss, keeping all other settings unchanged.

1) Effects of Vision Instruction Representation: The addition of S-PAE consistently improves retrieval performance. Compared with the *baseline*, +*S-PAE* yields a modest increase in mR, accompanied by a clear gain in T2I R@1. When combined with T-PAE, S-PAE further raises mR (36.95 → 37.25), with notable improvements in I2T and T2I R@10. Incorporating affiliation loss amplifies these benefits: +*S-PAE*+*T-PAE*+$\mathcal{L}_a$ outperforms +*T-PAE*+$\mathcal{L}_a$ by 2.6% in mR, with I2T R@1 increasing from 17.04 to 18.14 and T2I R@1 from 11.81 to 12.17.

For PriorCLIP-OD, S-PAE improves mR from 45.35 to 48.02 relative to *CLIP-baseline*, and from 48.16 to 52.68

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Fig. 6. Visualization of PirorCLIP, RemoteCLIP, and GeoRSCLIP on the top 5 retrieval results includes (left) image–query–text and (right) text-query-image. Blue text and boxes represent the queries, green text and boxes represent the matching retrieval results, and red text and boxes indicate the mismatched retrieval results (both fine-tuned on RSICD and RSICD).

when combined with $\mathcal{L}_a$. These results confirm that S-PAE significantly strengthens vision representation and enhances T2I retrieval.

2) Effects of Language Cycle Attention: T-PAE provides substantial improvements in language representation. Relative to the *baseline*, *+T-PAE* increases I2T R@1 and T2I R@1 by 24.6% and 35.8%, respectively. Adding T-PAE to S-PAE yields an additional 1.2% improvement in mR, while incorporating it into *+S-PAE*+$\mathcal{L}_a$ results in a further 1.4% mR gain (I2T R@1: 16.81 → 18.14; T2I R@1: 11.59 → 12.17).

Consistent improvements across multiple configurations demonstrate that T-PAE complements S-PAE and strengthens retrieval in both I2T and T2I tasks.

3) Effects of Affiliation Loss: The affiliation loss $\mathcal{L}_a$ also contributes significantly. Compared with the *baseline*, +$\mathcal{L}_a$ increases I2T R@1 from 16.15 to 19.25. When applied together with S-PAE and T-PAE, $\mathcal{L}_a$ further raises mR by 2.7% (I2T R@1: 17.04 → 18.14).

In the open-domain setting, $\mathcal{L}_a$ improves mR from 45.35 to 48.16 relative to *CLIP-baseline*, and from 48.02 to 52.68 when combined with S-PAE. These results indicate that affiliation loss not only enhances vision and language representations but also enables S-PAE and T-PAE to capture long-range dependencies better.

F. Parameter Evaluation

1) Filter Size in PriorCLIP: In PriorCLIP with the hard-belief strategy, the visual features extracted by the Transformer often contain semantic noise. In contrast, only a subset of these features is highly representative of the underlying visual semantics. To mitigate the effect of noise, we regulate the number of key features by adjusting the filter size parameter. The Vision Transformer outputs a maximum of 50 local features, from which a subset is selected.

Table IV reports the bidirectional retrieval performance on the RSITMD dataset under different filter sizes. When the filter size is set to ten, I2T retrieval achieves the highest R@10 of 54.65. At a filter size of 20, mean Recall reaches 38.00, while T2I R@1 attains its maximum of 13.50. A filter size of 30 again benefits I2T R10 but yields relatively lower mR. The

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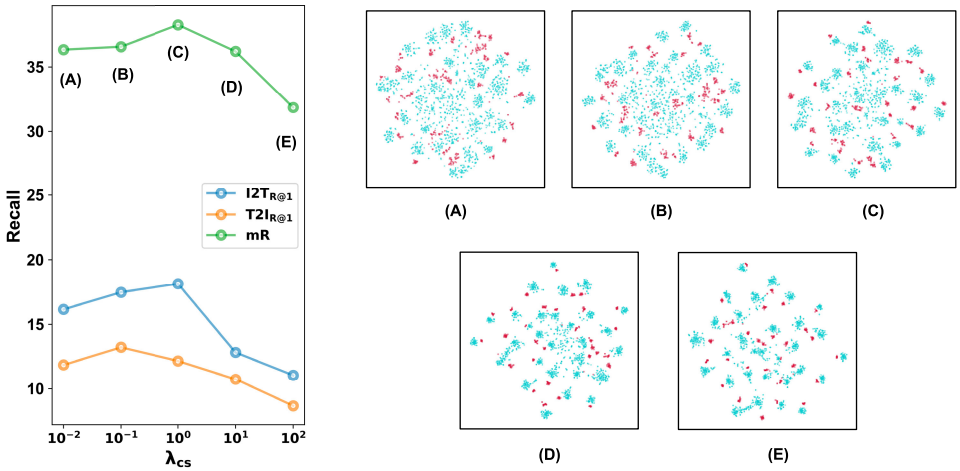


Fig. 7. Effect of (left) center scale on retrieval results and (right) RSITMD test set visualized with t-SNE [84] in the embedding space, the red squares represent vision embeddings, and the blue triangles represent text embeddings.

TABLE IV

EFFECT OF FILTER SIZE IN S-PAE ON THE RSITMD TEST SET. BEST RESULTS ARE GIVEN IN BOLD, AND SECOND-BEST RESULTS ARE GIVEN IN UNDERLINED

Filter Size (r_k)	Image→Text R@1 / R@5 / R@10	Text→Image R@1 / R@5 / R@10	**mR**
10 (hard)	17.26 / 40.71 / 54.65	13.01 / 41.06 / 61.19	37.98
20 (hard)	17.48 / 39.60 / 52.21	**13.50** / 42.48 / 62.74	38.00
30 (hard)	**19.03** / 37.83 / 49.78	12.83 / 41.90 / 62.08	37.24
40 (hard)	18.14 / 41.15 / 52.88	12.17 / 41.68 / **63.41**	38.24
50 (hard)	18.36 / 37.61 / 50.88	13.10 / 42.88 / 62.04	37.48
50 (soft)	18.36 / **42.04** / **55.53**	13.36 / **44.47** / 61.73	**39.25**

TABLE V

OVERALL EFFICIENCY COMPARISON OF REMOTE SENSING CLIP MODELS ON RSITMD TEST SET

Model	Total Time (s) ↓	Total GFLOPs ↓	Total Params (M) ↓
RemoteCLIP	1.6493	9.7690	151
GeoRSCLIP	1.6142	9.7690	151
PriorCLIP	1.8981	9.8823	189

best balance is observed when the filter size is 40, achieving the highest mR of 38.24, suggesting that a moderate number of features captures the most representative visual cues. When the filter size is expanded to 50, mR declines to 37.48, indicating that including all features reintroduces semantic noise that suppresses the most informative ones.

In contrast, PriorCLIP with the soft-belief strategy avoids the need to fix the number of features. It dynamically assigns weights, reaching an mR of 39.25, which outperforms all fixed filter sizes. These findings highlight that semantic noise in remote sensing images tends to deactivate key features, and performance can be enhanced by reducing reliance on redundant or noisy visual features.

2) Center Scale of Affiliation Loss: The affiliation loss is designed to reduce semantic confusion in the shared embedding space by encouraging intraclass compactness. The strength of this constraint is governed by the center scale parameter, denoted as λ_{cs}. The left panel of Fig. 7 shows retrieval performance on RSITMD with varying center scales, while the right panel illustrates the corresponding embedding distributions.

As the center scale increases, the interclass separation becomes more pronounced, and semantic confusion zones gradually diminish, as illustrated by regions A–E in Fig. 7. However, retrieval accuracy first improves and then declines. The optimal performance is achieved when $\lambda_{cs} = 10^0$, which balances intraclass compactness with interclass separation. Extreme constraints may lead to over-clustering, thereby limiting generalization and reducing retrieval effectiveness.

TABLE VI

RESULTS OF EXPERIMENTS OF DIFFERENT INSTRUCTION STRATEGIES ON THE RSITMD TEST SET

Instruction Encoder	**Pre-trained Dataset**	**Image-query-Text** R@1 / R@5 / R@10	**Text-query-Image** R@1 / R@5 / R@10	**mR**
VGG-16	AID	16.15 / 37.61 / 52.21	11.77 / 40.88 / 61.95	36.76
VGG-19	AID	**19.47** / 38.94 / 53.10	12.52 / 39.91 / 61.28	37.54
ResNet-50	AID	18.14 / **41.15** / 52.88	12.17 / 41.68 / **63.41**	38.24
ResNet-101	AID	17.26 / 39.38 / 52.88	11.81 / 41.02 / 61.55	37.32
ViT	AID	15.71 / 37.39 / 51.11	12.43 / 41.73 / 60.18	36.42
Swin-T	AID	15.93 / 36.28 / 53.32	12.65 / 41.68 / 60.84	36.78
VGG-16	ImageNet	16.81 / 36.28 / 49.12	12.70 / 41.28 / 62.04	36.37
ResNet-50	ImageNet	16.81 / 38.27 / 52.21	**13.81** / 41.68 / 61.64	37.40
Swin-T	ImageNet	15.71 / 40.93 / 52.65	13.01 / 40.31 / 60.18	37.13
VGG-16	RESISC45	16.81 / 38.27 / 50.66	11.50 / 40.58 / 60.31	36.36
ResNet-50	RESISC45	17.70 / 39.82 / **53.98**	12.92 / **42.96** / 62.30	**38.28**
Swin-T	RESISC45	15.93 / 37.61 / 52.21	12.08 / 41.06 / 61.37	36.71

G. Instruction Strategy Analysis

To investigate the role of instruction encoders, we evaluate PriorCLIP with different encoder architectures under the hard-belief strategy. Each encoder is pretrained on classification tasks using the AID [69], RESISC45 [85], or ImageNet [86] datasets. The candidate encoders are divided into two categories: 1) CNN-based architectures, including VGG-16 [87], VGG-19 [87], ResNet-50 [68], and ResNet-101 [68]; and 2) Transformer-based architectures, ViT [26] and Swin-T [83].

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Experimental results on the RSITMD dataset are presented in Table VI. Among all configurations, ResNet-50 consistently achieves the highest performance, with mean Recall (mR) values of 38.24 and 38.28 when pretrained on AID and RESISC45, respectively. In contrast, Transformer-based encoders such as ViT and Swin-T provide no significant improvements when pretrained on AID, suggesting that Transformer-based priors are less effective for this task. A comparison between VGG-16 and ResNet-50 trained on ImageNet further indicates that pretraining on remote sensing scene classification datasets provides stronger and more relevant prior knowledge than pretraining on generic natural images. Because AID and RESISC45 are both remote sensing datasets, their prior information is more useful for guiding visual representation. Remote sensing images share similar texture patterns and spatial structures, so models can transfer knowledge more effectively across these datasets. In terms of architecture, ResNet and VGG extract features using local convolutional kernels, which makes them good at capturing textures and small spatial details in remote sensing images. In contrast, ViT relies on global self-attention and usually needs larger training datasets to perform well. Therefore, on small or medium-sized remote sensing datasets, CNNs like ResNet and VGG often show more stable performance than ViT.

Overall, these findings demonstrate that instruction strategies derived from remote sensing scene recognition deliver more effective priors for guiding visual representations. Leveraging such domain-specific knowledge enables PriorCLIP to produce more reliable embeddings and, consequently, improved retrieval performance.

VI. Conclusion

In this work, we proposed the visual-prior-guided vision–language model, a novel paradigm that leverages prior knowledge from the remote sensing domain to enable adaptive learning of vision and text representations. Within this model, we developed PriorCLIP for RSITR. Specifically, S-PAE and T-PAE were designed to model long-range dependencies and enhance key feature representations. For vision representation, prior knowledge from remote sensing scene recognition was utilized to construct unbiased features and suppress semantic noise. For text representation, T-PAE recurrently activated the current time step using the previous step, thereby strengthening semantic encoding. Furthermore, we introduced a vision-instruction fine-tuning strategy to mitigate semantic noise in remote sensing vision–language representations, enabling more robust open-domain retrieval. In addition, a clusterwise Affiliation loss was proposed to enforce interclass constraints and reduce semantic confusion in the shared subspace. Extensive experiments on RSICD and RSITMD verified the superiority and effectiveness of PriorCLIP, consistently achieving state-of-the-art performance in both closed-domain and open-domain scenarios.

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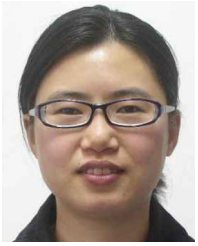
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